



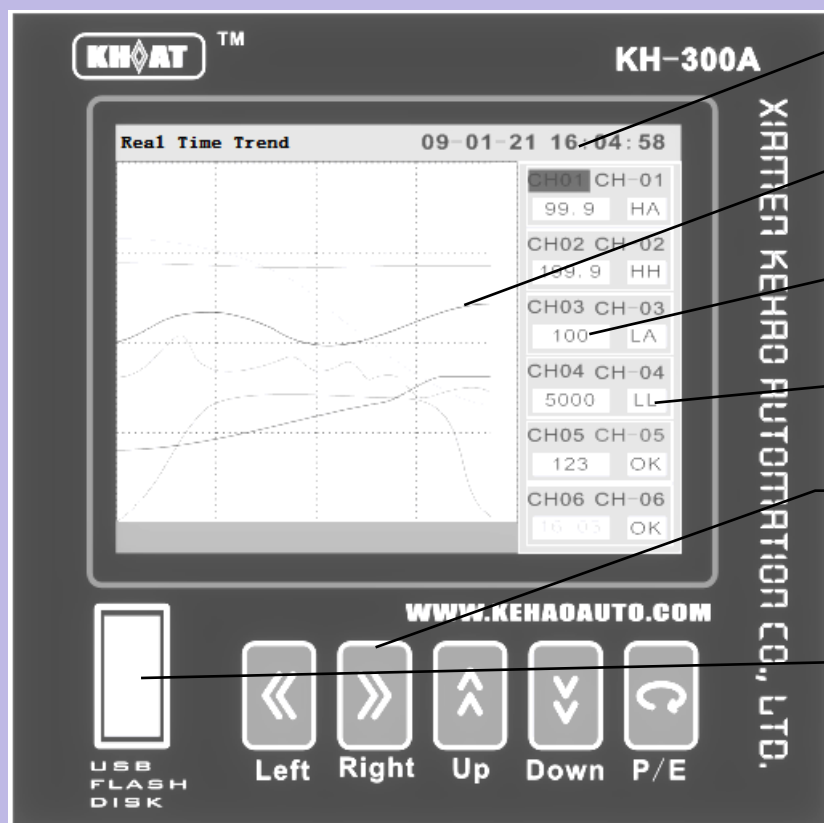
KH300AG Intelligent Color Paperless Recorder

KEHAO INSTRUMENT



Memory Flash Drive Data Saving

Provides flexibility and variety in the handing of record data



Time Display

Indicate date and time of recorded data

Trend Display

Allow you to view data in trend forms

Digital Display

Allow you to view data in digital form

Alarm State

Alarm state display: HH,LL,H,L when alarm

Key Panel

Allows you to perform selection of display, setting, data change, and channel no. change

USB Port

Allows you to download data from recorder to PC by USB flash drive

Long Time Memory

8MB built in memory for long time memory. Record time = 45 days ÷ Channel No.s x Record interval time

Universal Input, 6 Points Max. Recording

Photoelectrical isolated universal input, 7 types of thermocouples, 3 types of resistance bulb, DC voltage, DC current are available.

USB Download Data Directly

Saved data transferred by USB flash drive directly, plug and play, easy to operation

PC Support Software

Data Analysis Software supplied in a CD-ROM as part of accessory. DCS configurable software as option.

Screensaver

Recorder turns off the backlight of LCD and exit parameter setting screen when non-operation.

Communication

Photoelectrical isolated RS485 serial communication is available. Standard MODBUS-RUT protocol with reading and writing functions.

Printing

Curve or digital data can be printed by connecting mini printer directly or through software

PC Support Software

Display data in digit and trend format, Export data as Excel format for further analysis, Print trend data by office printer.

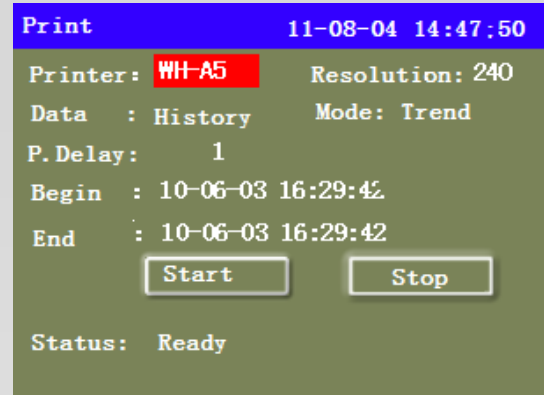
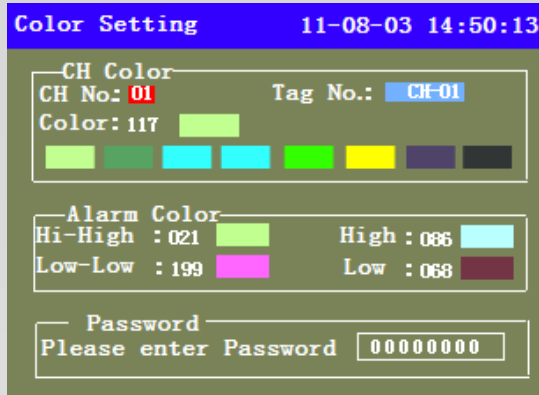
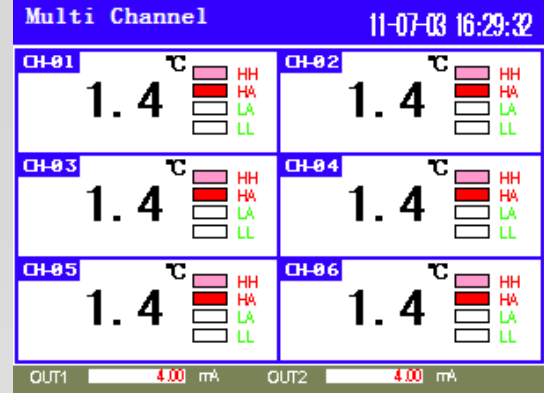
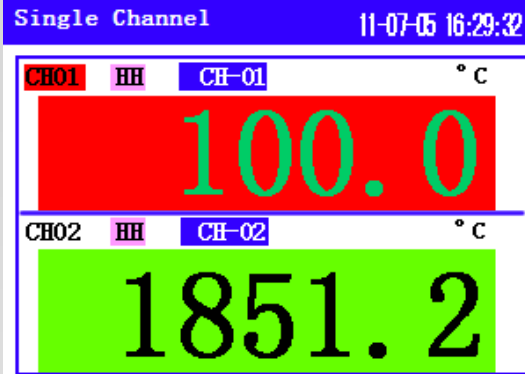
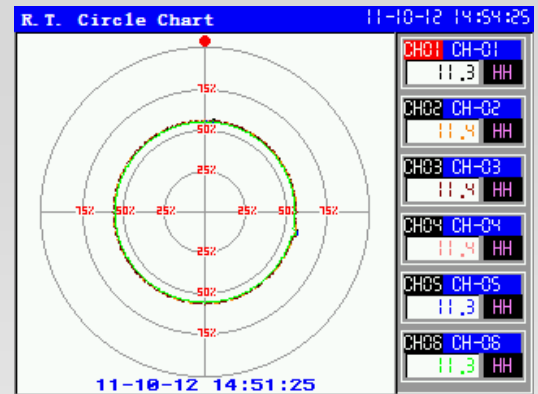
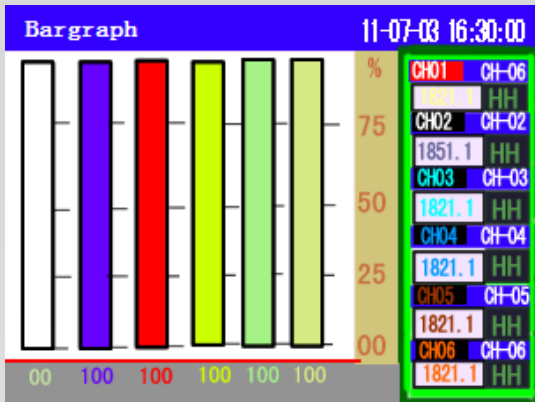
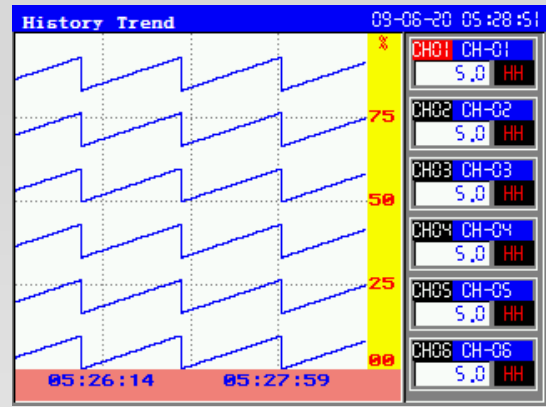
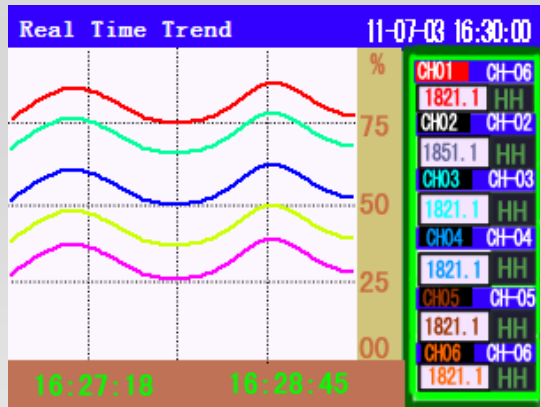
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ALP-LAB, napredne tehnologije, d.o.o.

Vošnjakova ul. 12, 2250 Ptuj, Slovenija

Web: www.alp-lab.si Email: info@alp-lab.si Tel: +386 (0)41 569 371

Various Screen Display Modes



Specifications

General Specification

Power Supply	100-240VAC, 47-63Hz; 24VDC
Power Consumption	Maximum 5VA(5W)
Insulation	Power to ground > 1500VAC Power to housing > 1500VAC
Keyboard	P/E, Left, Right, UP, Down
Materials	ABS for case and bezel
Terminal	M5 screw terminal
CPU	32bits, high performance and integrated ARM
Mount Method	Panel Flush Mounted
Size/Mount Size/Mass	96x 96x85mm, 92x92mm, 0.5kg
Operation Temperature	Working Temperature:0-50C Relative humidity; 10%-85%(now dew)
Transport /Storage	Temperature: -20-60℃ Relative humidity : 5%-95%(nodew)

Input Specification

Number of inputs	1, 2, 3, 4,5,6 points
Input Signal	Thermocouple: 7 types (K,S,B,E,J,N,T) RTD-Resistance bulbs: 3 types (Pt100, CU50,CU100) DC Voltage: (0-5VDC, 1-5VDC) DC Current: (4-20mA, 0-10 mA) Others: 4 types (0-20mV,0-60mV, 0-100mV, 0-500mV)
Sample Rate	1s/6 channels,
Scaling	-20000 to 20000
CMR Ratio	85-110dB
Temperature Shift	50PPM
Photoelectrical Isolation	1000VAC between channels ground 400VAC between channels
Input Independence	0-5VDC and 1-5VDC input: 500KΩ 4-20mA input: 250 KΩ 0-10mA input: 500KΩ Other signal input :20MΩ
Input Error Action	Max, Min, Hold

Display Specification

Display	3.2 inch, 320X240 TFT, color LCD display
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Life of Backlight	50,000hours
Screensaver Time	0-30000s, settable
Unit	77 , settable and addable each channel
Display Contents	- Real time Trend Display In horizontal, selected in the refreshment cycles of 1 to 3600sec Scale display - Real time Circular Chart Display - Digital display Single and multi channels display - Bargraph Display - History Trend Display - History Circular Chart Display - Printing Setting Display - System Configuration

Recording Specification

Memory Media	USB memory (2GB), FAT16 format
Memory	Flash memory
Memory Capacity	8MB built in for long time record
Record Interval	1 to 3600 seconds, settable flexibly
Record Time	45days ÷ Channel no * Record interval 45days per 1s record interval per CH 7years per 1 min. record interval per ch
Recording Method	Start recording when power on. Stop recording when power off.
Data Save Cycle	Oldest data replaced by newest data accordingly when memory is full
Data Format	Binary format or cannot read or write

Optional Output Function

Relay Alarm	up to 2 points, 250VAC,3A, NO or NC
Retransmission	4-20mA, up to 2 point
Print	RS232 print port
USB	USB flash drive
Auxiliary Power Supply	24VDC, 50mA for sensor and transmitters
Communication	RS485 MODBUS-RTU protocol

Alarm Function

Type of Alarm	High, Low, High-High, Low-Low limit
No. of Setting	Up to 4 alarms are settable for each channel, configurable flexibly

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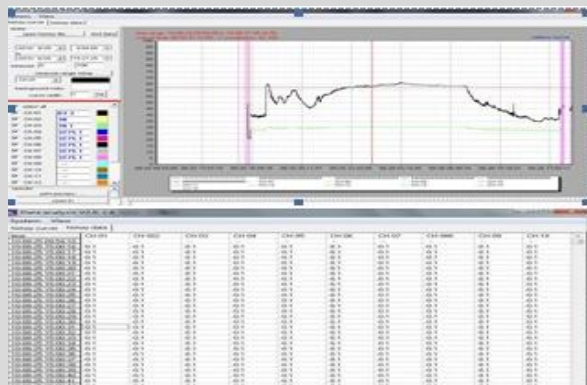
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Specification

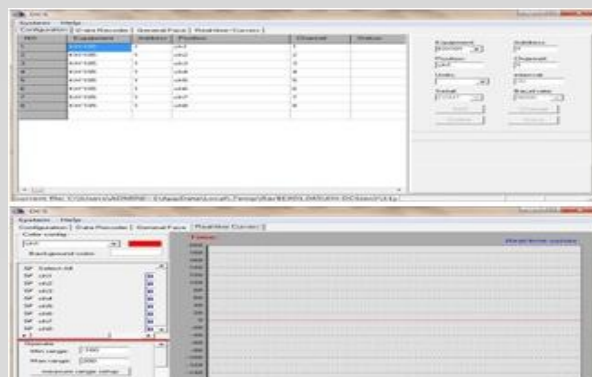
Alarm Indication	Alarm States is displayed in digital, trend, bargraph, circular chart. When alarm occurs, state flashing	Clock	Clock accuracy: +5ppm. After power off, Li battery for continual power supply. The validity of battery is 30days.
Alarm output	Up to 2 points, 3A contact	PC Support Software (Standard-supplied CD-ROM)	
Alarm Setting	Individual or common output	O/S	Window 2000/XP, WISTA
Communication Function		Required Hard Disk Capacity	Free capacity of 30MB or larger required
Communication	Photoelectrical isolated RS485 communication port, read and write	Required Memory	1GB or larger
Protocol	Modbus RTU, can communicate with modern PLC and HMI directly	Contents	The follows types are included as standard: 1) Data Analysis Software • It allows you to view the past recorded data in digit and curve format from data saved in recorder to USB flash drive • It allows you to export the data as Excel format for further analysis • It allows you to print the curve data by office printer 2) DCS configuration Software • It allows you to view real time data in digit, curve format • It can save the past data with same function with data analysis software.
Cable	RS485 shielded twisted pair cable		
Print Function			
Print Port	RS232C comm. Port, Baudrate: 9600		
Printer	Dot-matrix mini printer; Ribbon Resolution: 60,120,240dots/line		
Data Printed Type	History data, Real time data , history curve data, optional		
Reference Performance			
Accuracy	0.2 grade when RTD, linear voltage, linear current and T.C input 0.2%FS±2.0℃ when T.C input with cold junction compensation by internal part of recorder		
Indication Resolution	0.1℃		
Input Resistance	RTD: Current 2.5mA, three wire, max.10ohm per each wire. Thermocouple: not more than 1000Ω.		
C.J Compensation	Tolerance: maximum +2℃		

PC Support Software

History data transferred to USB flash drive can be viewed to personal computer



History and real time data by RS485 can be viewed to personal computer.



PC Support Software

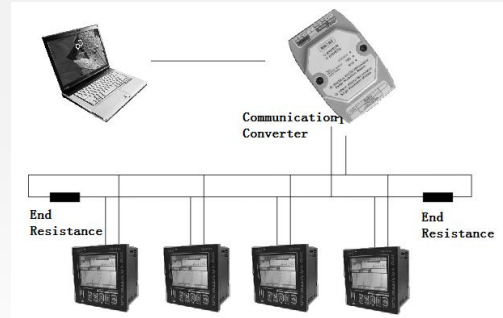
USB Download Data, Plug and Play



Please install the software in CD and USB flash drive to PC before usage.

- O/S: Window 2000/XP, WISTA
- Required hard disk capacity :Free capacity of 30MB or larger required
- Recommend USB flash driver: Kingston
- FAT16 format.

Communication

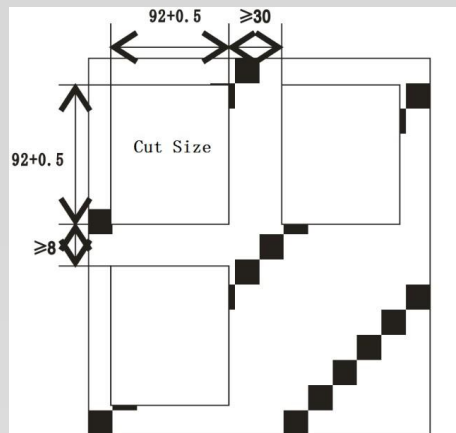
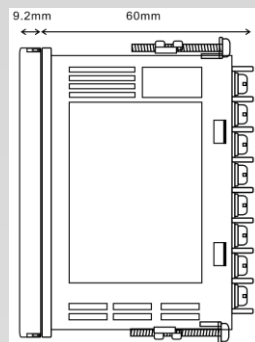
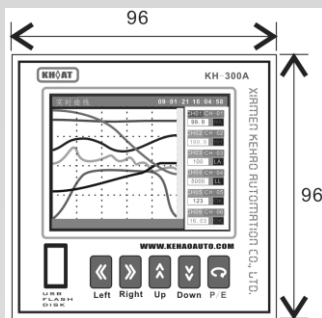


Please install the software in CD before usage.

- O/S: Window 2000/XP, WISTA
- Required hard disk capacity :Free capacity of 30MB or larger required
- Recommend cable: RS485 shielded twisted pair cable
- Repeater is needed when more than 100m communication distance.

Size & Installation (Unit: mm)

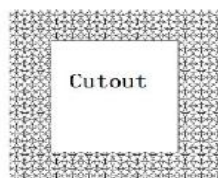
Dimension



Installation

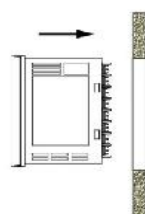


Mounting Panel



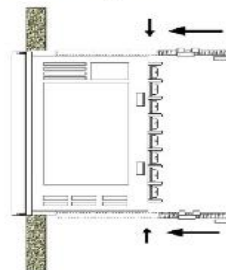
1. Cutout

Mounting Panel



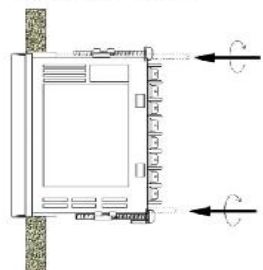
2. Mounting Meter

Mounting Panel



3. Mounting Bracket

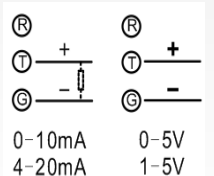
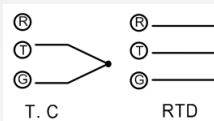
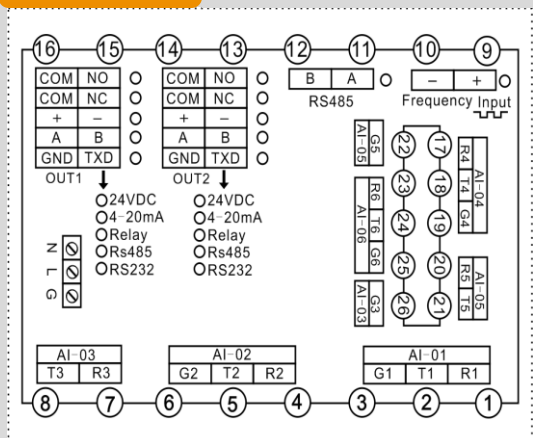
Mounting Panel



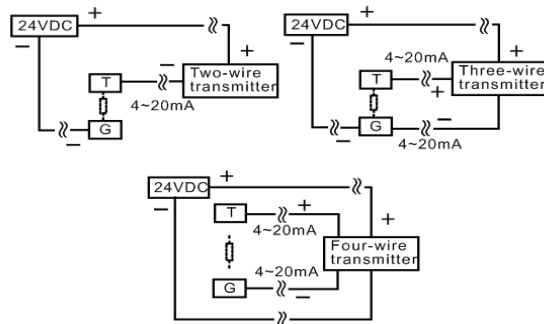
4. Fix Bracket

Diagram & Connection (Unit: mm)

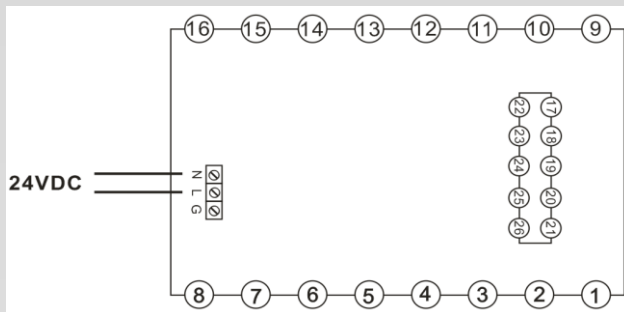
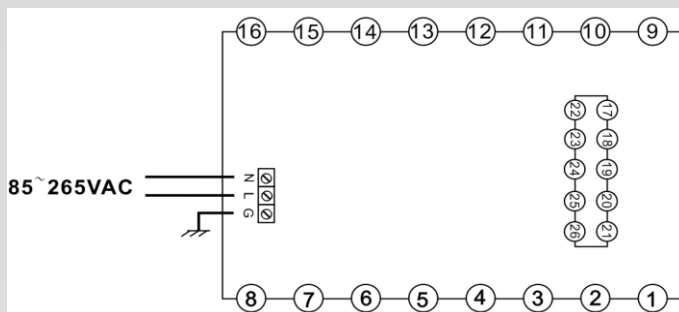
Diagram



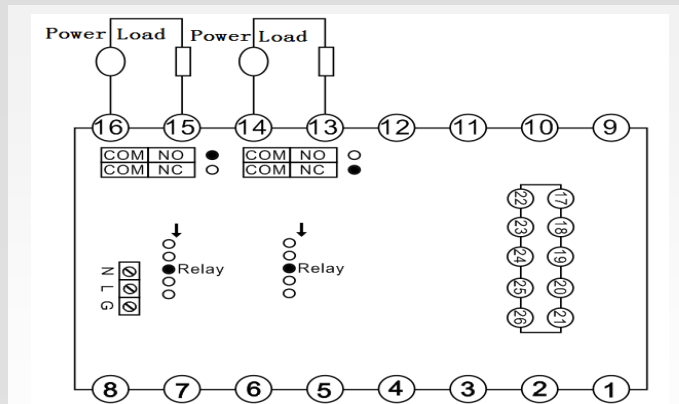
TRANSMITTER



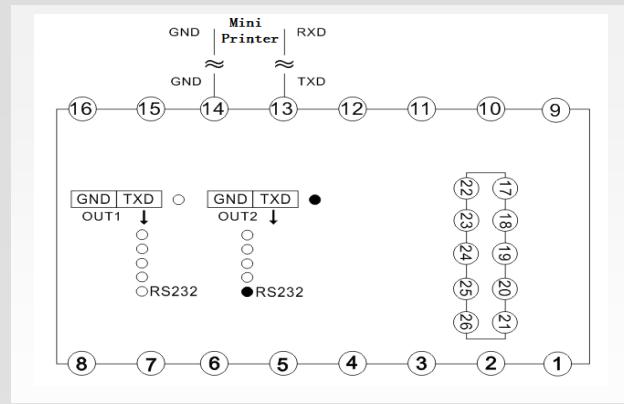
Power Diagram



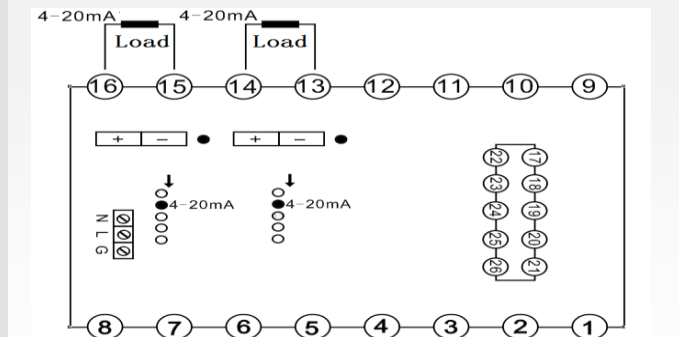
Alarm Output Diagram



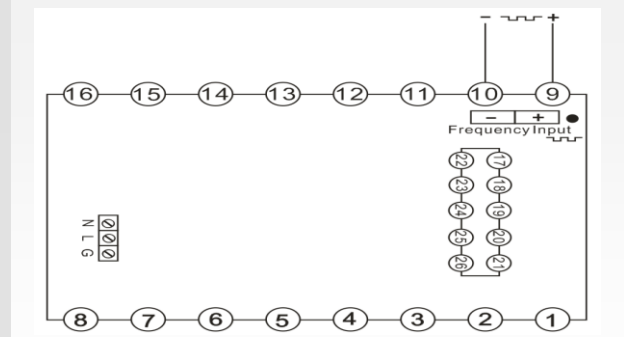
Mini Printer Diagram



4-20mA Output Diagram



Frequency Input Diagram



Order Code

Function	Code and Description										
Basic Code	KH 3	A					U-				KH300 Paperless Recorder
Channel No.	01										One Channel
	02										Two Channels
	...										
	06										Six Channels
Size	A										96*96*85mm(L*W)
LCD color	G										Color
OUT1	-N										None
	-R2A										Relay alarm: NO ,30VDC/3A, 220VAC/3A
	-R2B										Relay alarm: NC ,30VDC/3A, 220VAC/3A
	-U3										Isolated auxiliary 24VDC power supply for transmitter, sensor and other device, max.50mA
	-I1										Isolated4-20mA retransmission output
OUT2	-N										None
	-R2A										Relay alarm: NO ,30VDC/3A, 220VAC/3A
	-R2B										Relay alarm: NC ,30VDC/3A, 220VAC/3A
	-U3										Isolated auxiliary 24VDC power supply for transmitter, sensor and other device, max.100mA
	-P										RS232 printing port for mini printer, WH-A5 mini printer as default.
	-I1										Isolated 4-20mA retransmission output
Communication	-N										None
	-S1										RS485 communication port
USB	-U										USB flash drive for download data
Frequency Input	-N										None
	-Q										One channel, 0-5KHZ frequency input
Power Supply	-N										220VAC, 50HZ ,85-240VAC
	-A										110VAC, 60HZ,85-240VAC
	-D										24VDC
PC Support Software	-N										Free data analysis software for USB to PC, no communication application
	-E										Extensive DCS software RS485 communication
Mini Printer	-N										None
	-W										Yes, Kehao mini dot-matrix printer

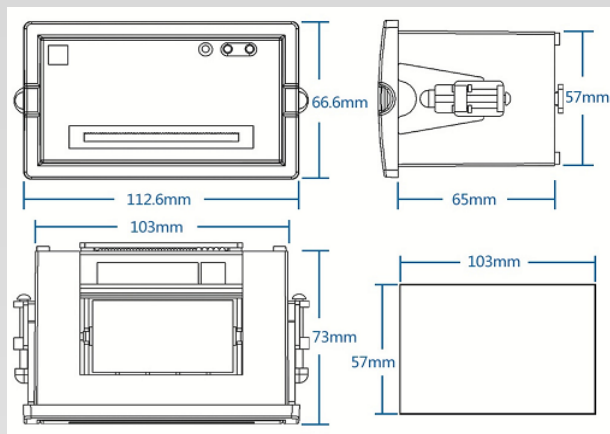
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Optional-Mini Printer

Printer



Size



Printing Method	Dot Matrix, Ribbon Type	Power Supply	5VDC,1.5A
Paper Width	44mm/57mm	Outline Size:	122.6x66.6x73mm
Print Width	32mm/48mm	Cut Size	103mm(W)x57mm(H)x65mm(D)
Resolution	96dots/line, 144dots/line,240dots/line	Interface	Serial RS232port
Character per line	16/24/40	Operate Temperature	0-5C
Printing Speed	1line/sec, 0.7 line/sec, 0.4line/sec.	Storage Temperature	-20 to 60C
Character Size	6x8dots, 8x16dots,6x 12dots	Operating Humidity	10%-85%
Paper Type	Plain, 44mm/57mm wide, 30mmΦ	Storage Humidity	10%-90%

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Email: admin@khmeter.com.cn cindysun2011@gmail.com

ATTN: Cindy Sun Sales Manager

Website: www.kehaoauto.com/en

Caution on Safety

Read instruction manual before using the product

Information in the catalog is subject to change without notice.

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